

Guangzhao (Alex) He

www.guangzhaohe.com — gh466@cornell.edu

Education

Cornell University

Ph.D. Computer Science

Advisors: **Prof. Hadar Elor** and **Prof. Wei-Chiu Ma**

Ithaca, NY, US

Aug. 2025 — Present

Zhejiang University

Chu Kochen Honors College

B.Sc. Computer Science and Technology

GPA: 3.99/4.0

Hangzhou, China

Aug. 2021 — Jun. 2025

Publications

NeuROK: Generative 4D Neural Object Kinematics

Chen Geng*, **Guangzhao He***, Yue Gao*, Yunzhi Zhang, Shangzhe Wu and Jiajun Wu

Under Review, 2026

Category-Agnostic Neural Object Rigging

Guangzhao He*, Chen Geng*, Shangzhe Wu and Jiajun Wu

CVPR 2025

ERNet: Efficient Non-Rigid Registration Network for Point Sequences

Guangzhao He, Yuxi Xiao, Sida Peng, Zhen Xu and Xiaowei Zhou

ICCV 2025

4k4d: Real-time 4d view synthesis at 4k resolution

Zhen Xu, Sida Peng, Haotong Lin, **Guangzhao He**, Jiaming Sun, Yujun Shen, Hujun Bao, and Xiaowei Zhou

CVPR 2024

EasyVolcap: Accelerating Neural Volumetric Video Research

Zhen Xu, Tao Xie, Sida Peng, Haotong Lin, Qing Shuai, Zhiyuan Yu, **Guangzhao He**, Jiaming Sun, Hujun Bao, and Xiaowei Zhou

SIGGRAPH Asia 2023 TC

Research Experience

Evaluating Motion Persistence in Video Models

Aug. 2025 — Present

Advised by Prof. Hadar Elor and Prof. Wei-Chiu Ma, Cornell University

- Engineered a data generation pipeline utilizing 360° videos to produce perspective video pairs representing multiple camera view sequences of the same scene.
- Constructed a benchmark dataset designed to quantify object permanence and motion persistence in video models.
- Developed a comprehensive testing framework and automated metric pipeline to benchmark performance across various video reconstruction and generation models.

Implicit Neural Object Kinematics

Feb. 2025 — Dec. 2025

Collaboration with Prof. Jiajun Wu at Stanford Vision and Learning Lab, Stanford University

- Created a large-scale 3D/4D assets dataset for learning generic kinematics of category-agnostic objects.
- Trained a latent pose-conditioned shape generative model that serves as a universal kinematics representation.

Category-agnostic Neural Rigging Representation

Jul. 2024 — Jan. 2025

Research Intern with Prof. Jiajun Wu at Stanford Vision and Learning Lab, Stanford University

- Proposed a category-agnostic neural rigging representation using embedded anisotropic blobs, which supports interactive pose editing of any object category, trained end-to-end in a data-driven manner.
- Pioneered an implicit, data-driven approach for object skinning, eliminating artifacts from traditional methods like LBS.
- Developed an architecture that makes feed-forward blob prediction and re-posed shape reconstruction in an auto-encoding fashion.
- Achieved unsupervised identity and pose disentanglement by proposing a dual-encoder architecture.

Generalizable Non-rigid 4D Registration via Deformation Graph Estimation

Dec. 2023 — May 2024

Research Assistant with Prof. Xiaowei Zhou at State Key Lab of CAG&CG, Zhejiang University

- Designed and implemented a fast, accurate, and generalizable architecture for non-rigid point sequence registration, achieving over 5 FPS, 4x faster than previous methods, while doubling accuracy.
- Introduced a novel approach for predicting deformation graphs using a custom spatio-temporal Transformer in a feed-forward manner.
- Improved temporal consistency and accuracy by employing a coarse-to-fine strategy and a two-stage training process.

Real-Time Novel View Synthesis for Dynamic Scenes

Mar. 2023 — Nov. 2023

Research Assistant with **Prof. Xiaowei Zhou** at State Key Lab of CAG&CG, Zhejiang University

- Implemented a novel, differentiable point-based rendering engine that supports PyTorch differentiation through OpenGL depth peeling, achieving fast rasterization with 80+ FPS under 4k resolution on a single RTX 4090 GPU.
- Explored both 3DGS-based and point-based representations with addition of SDF.
- Augmented real-time dynamic scene rendering pipeline with both Nerf-Acc and point-based efficient sampling to achieve 2x improvement in rendering speed without compromising quality.

Universal Framework for Neural Volumetric Video (EasyVolCap)

Mar. 2023 — Jul. 2023

Research Assistant with **Prof. Xiaowei Zhou** at State Key Lab of CAG&CG, Zhejiang University

- Proposed a universal framework for neural volumetric video research, including a powerful configuration system, common network backbones, multiple rendering backends and a streaming visualizer.
- Implemented multiple works based on the framework, including Instant-NGP, D-NeRF, E-NeRF and K-Planes.

Awards

Young Scientists Fund of the National Natural Science Foundation of China

Sep. 2024 — Jul. 2025

Funded for research on generalizable 4D reconstruction, top 0.1%.

Top 10 Outstanding Students at Chu Kochen Honors College, Zhejiang University

Dec. 2024

Awarded for academic and innovative outstanding, top 0.5%.

National Student Research Innovation Competition — 1st Place

Oct. 2024

Awarded grand prize to top 2 contestants nation-wide for excellence in research and innovation.

Zhejiang Provincial Government Scholarship

Nov. 2022

Awarded for academic and innovative excellence, top 5%.

Academic Excellence Award, Zhejiang University

Oct. 2022, 2023, 2024

Awarded for academic excellence, top 15%.

Project Experience

RV32I CPU. Designed and implemented a 5-stage pipelined RISC-V CPU with RV32I instruction set support and partial RV32F support, featuring a custom ALU for floating-point operations, using Vivado, Verilog, and FPGA.

C Compiler for RISC-V64. Developed a 15k-line C compiler for RISC-V64 platforms, supporting essential C syntax and grammar, with features including lexical and semantic analysis, optimized intermediate representation, and efficient assembly code generation using the maximal munch algorithm and graph coloring.

Linux for RISC-V64. Built a Linux-like operating system for RISC-V64 devices using C and RV64 assembly, featuring thread scheduling, virtual memory management, exception handling, and user-mode program execution, integrated with a forking mechanism and compatibility with code generated by the custom C compiler.

MiniSQL Database Management System. Created a 20k-line MiniSQL DBMS in C++ with a Buffer Pool Manager using bitmap and LRU policies, a Record Manager with table heap support, and an Index Manager implemented with a B+ Tree structure.

Skills

- **Language:** TOEFL 109, GRE 323, IELTS 7.5
- **Programming:** Python, C++, C, Java, MATLAB, Dart, HTML, $\text{\LaTeX}$
- **Research Tools:** PyTorch, NumPy, OpenGL, OpenCV
- **Signed Music Producer:** 40+ singles with more than 2 million streams